

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
Higher 2

CANDIDATE  
NAME

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CLASS

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TUTOR'S  
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## BIOLOGY

**9747/03**

Paper 3 Longer Structured Questions

**18 September 2008**

**1 hour 30 minutes**

Candidates answer on the Question Paper.

No Additional Materials are required

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### READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions.

#### Section B

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| 1                  |  |
| 2                  |  |
| 3                  |  |
| 4                  |  |
| Total              |  |

### Section A

Answer all questions in this section

1. Restriction enzymes are used during genetic engineering. The target sites for *EcoRI*, *BamHI* and *AluI* is shown in the table below.

|              |                                            |
|--------------|--------------------------------------------|
| <i>EcoRI</i> | 5' ...G↓AATTC ...3'<br>3' ...CTTAA↑G ...5' |
| <i>BamHI</i> | 5' ...G↓GATCC ...3'<br>3' ...CCTAG↑G ...5' |
| <i>AluI</i>  | 5' ...AG↓CT...3'<br>3' ...TC↑GA...5'       |

Table 1.1

A bacterial plasmid is shown in the figure below. There are no target sequences for the three restriction enzymes in the rest of the plasmid.

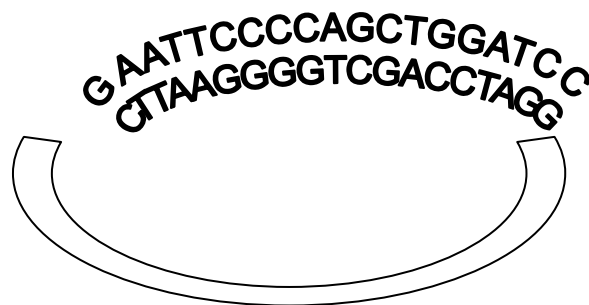


Fig 1.1

With reference to **Table 1.1** given above,

- (a) State how many fragments of DNA will be present after the plasmid has been cut with

(i) a mixture of *EcoRI* and *BamHI* [1]

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(ii) a mixture of *EcoRI* [1]

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- (b) Explain why it is more suitable to use *EcoRI* than *AluI* in genetic engineering of the plasmid.[2]

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- (c) A student incubates the plasmid molecule shown in Figure 1.1 with *EcoRI* to insert a human gene.

Outline how restriction enzymes may be used in the formation of recombinant DNA molecules. [3]

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- (d) Name two other enzymes used in the formation of recombinant DNA and briefly describe their roles. [2]

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- (e) Gel electrophoresis is usually performed after restriction digestion to separate DNA fragment. After which, nucleic acid hybridization is conducted.

(i) Briefly describe how DNA fragments may be separated by gel electrophoresis.[3]

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(ii) Describe how the process of nucleic acid hybridization is carried out. [4]

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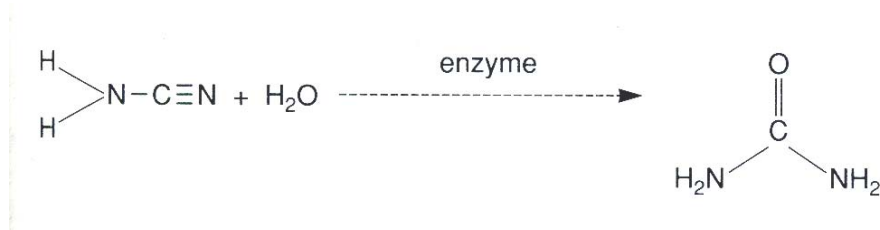
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[Total: 16 marks]

2. Read the following passage about genetic engineering and answer the questions that follow.

A 5% cyanamide kills the weeds which infest fields of tobacco plants, but causes some damage to the tobacco. However, tobacco plants have been genetically modified by inserting a gene from a soil fungus, *Myrothecium verrucaria*. The gene codes for an enzyme, cyanamide hydratase, which converts cyanamide into urea:



The gene was inserted into a plant virus and then tobacco plants were infected with the virus. A strain of transgenic tobacco plants was thus developed that carried the gene cyanamide hydratase but were unaffected by other viral proteins. The enzyme was most active in the leaves, showing about one-fifth of the activity found in *Myrothecium*.

The transgenic plants thrive when treated with a dilute solution of cyanamide, since the plants have urease enzymes which convert urea to ammonium ions which provide the plants with a source of nitrogen.

- (a) Explain why the tobacco plants described in the passage are transgenic plants. [1]

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- (b) Suggest why the enzymes activity in the transgenic plant is lower than the fungus from which the gene was taken. [2]

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- (c) Outline the possible benefits and potential hazards of this particular example of a genetically modified organism. [4]

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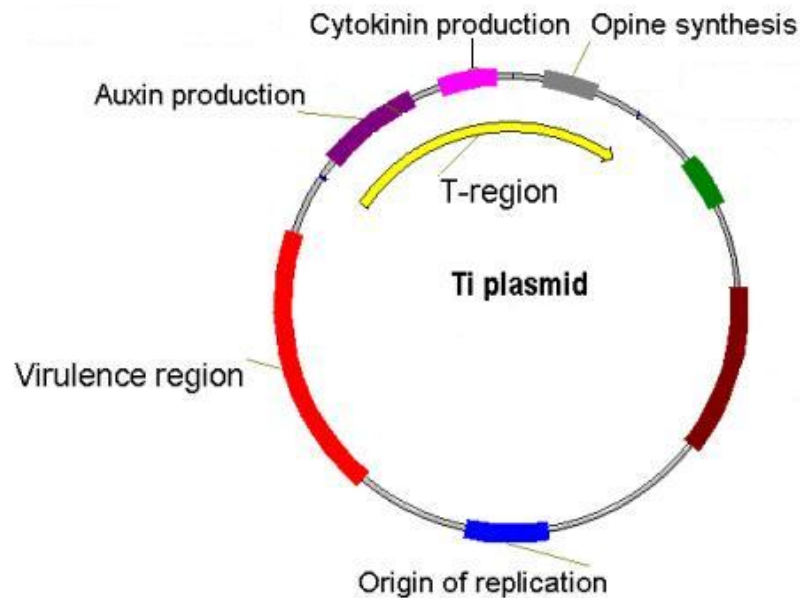
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The use of viruses as a vector to deliver the cyanamide hydratase gene is one method of delivering the transgene to the plant. Another commonly used method is the use of soil bacterium *Agrobacterium tumefaciens* as a vector.

- (d) State two other methods which can be used to deliver foreign genes to plants. [1]



**Fig 2.1**

- (e) The Ti plasmid in *Agrobacterium tumefaciens* has to be attenuated and modified before it can be used as a vector for genetic engineering. With reference to **Fig 2.1**, explain what this means. [3]

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- (f) Describe how a genetically modified tobacco plant containing the cyanamide hydratase transgene can be produced using *Agrobacterium tumefaciens*-mediated leaf disc transformation. [4]

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**[Total: 15 marks]**

3. Every day when she wakes up, Lisa begins to deal with a condition she's known all her life - **cystic fibrosis**. Her parents clap her back and chest for at least 20 minutes to help clear her lungs of the thick mucus that sometimes makes it difficult for her to breathe.

At school, Lisa coughs a lot, so she keeps a box of tissues on her desk just in case she coughs up mucus and needs to spit it out. In gym class, she participates in sports but often gets tired easily. At lunchtime, Lisa takes pills to help her digest her food and get all the vitamins she needs, which helps her keep up her strength to deal with cystic fibrosis.

- (a) Explain what is the cause of cystic fibrosis [2]

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- (b) Describe a suitable treatment that can help Lisa in leading more or less a normal lifestyle. [4]

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- (c) Explain the problems of the suggested therapy and suggest why Lisa will have to repeat the treatment periodically. [4]

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Genetically-modified fish, which can grow up to 10 times faster than normal, could be cleared for human consumption within a year. A US firm - AF Protein - is developing the GM fish on Prince Edward Island, Canada.

The fish would become the first genetically modified animal cleared for consumption, and the technology involved could cut the cost of raising salmon and trout by half. The Massachusetts-based company has inserted two sets of fish genes into Atlantic salmon. The first set are growth hormone genes and the second, from a different fish, activate them.

As a result, at the age of 18 months, the salmon are five times the size of their unmodified siblings.

*Article taken from BBC News 18 May 1999*

(d) Discuss the impact of genetic modified salmon on fishermen in Canada. [4]

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**[Total: 14 marks]**

### **Section B**

Answer all questions in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

4. (a) Discuss the benefits and ethical concerns of the Human Genome Project. [10]
- (b) Explain the uses of RFLP analysis. [6]
- (c) Explain the properties, using named examples of the normal functions of stem cells. [4]

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